Problem 1: Strawberry Market

In a small town, the market for fresh strawberries operates perfectly competitively. The demand and supply for strawberries in this town are governed by the following equations:

$$\text{Demand Equation: } Q_D = 60 - 5P$$
$$\text{Supply Equation: } Q_S = -20 + 10P$$

where Q_D and Q_S are the quantity demanded and supplied respectively, and P is the price of strawberries per pound.

Use this information to answer the following questions on the next slides.

Problem 1: Questions (a)-(b)

(a) Determine the equilibrium price and quantity.

(b) Draw and label the market equilibrium price and quantity as well as consumer surplus and producer surplus.

Problem 1: Questions (c)-(d)

(c) Price Increase Scenario: Suppose a disease affects the strawberry crop, reducing the supply. The new supply equation is: $Q_S = -40 + 10P$. Calculate the new equilibrium price and quantity, and illustrate the changes on your graph.

(d) Price Elasticity of Demand: If the price of strawberries increases from \$4 to \$5, calculate the price elasticity of demand using the difference method.

(e) **Consumer Surplus and Producer Surplus:** Calculate the consumer surplus and producer surplus at the initial equilibrium.

Problem 2: Notebook Market

In a medium-sized city, the market for notebooks is perfectly competitive. The demand and supply for notebooks are described by the following equations:

$$\text{Demand Equation: } Q_D = 250 - 25P$$
$$\text{Supply Equation: } Q_S = -30 + 20P$$

where Q_D and Q_S are the quantity demanded and supplied respectively, and P is the price of a notebook in dollars.

Use this information to answer the following questions on the next slides.

Problem 2: Questions (a)-(b)

(a) Determine the equilibrium price and quantity.

(b) Draw and label the market equilibrium price and quantity as well as consumer surplus and producer surplus.

Problem 2: Questions (c)-(d)

(c) **Price Elasticity of Demand:** Calculate the price elasticity of demand using the difference method if the price decreases from \$8 to \$7.

(d) **Consumer Surplus and Producer Surplus:** Calculate the consumer surplus and producer surplus at the initial equilibrium.